AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MST129
Applied calculus

Final Examination
Fall Semester 2025/2026

Date:13 /Jan. / 2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2:00 Hours

Instructions:

- Total Marks: 50
- This exam consists of **two** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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Q-1: [5×2 marks] Choose the correct answer:

1.1: which of the following is the simplification of $\ln x^5 - \ln x^3$

a) x^2
b) $2\ln x$
c) $15\ln x$
d) None of the above

1.2: the solution of the equation $e^{-4x} = e^{-12}$ is:

a) 4
b) -4
c) 1/4
d) None of the above

1.3: $\int -\sin x + 3\cos(3x)dx$

a) $\cos x + \sin(3x) + c$
b) $-\cos(x) + \sin(3x) + c$
c) $-\cos(x) - 3\sin(3x) + c$
d) None of the above

1.4: The value of $\int_0^{\pi} 1 - \sin(x)\ dx$ is:

a) π
b) $\pi + 2$
c) $\pi - 2$
e) None of the above

1.5: If $y = e^{\cos(x)}$, then $\frac{dy}{dx} =$

a) $e^{\cos(x)}$
b) $\sin(x)(e^{\cos(x)})$
c) $(-\sin(x))e^{\cos(x)}$
d) None of the above

Each question is worth 10 marks. Answer the following questions:

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Q–2:

a) **[6 marks]** Let $f(x) = x^3$-12x $-3 \leq x \leq 5$
Find the absolute maximum and absolute minimum of f.

b) **[4marks]** Use implicit differentiation of the equation $y^4 + x^4 = 2$ *to* find the slope of the graph at the designated point; $P(1,1)$.

Q–3:

a) **[5 marks]** Find the equation of the tangent line to the curve $y = e^x + x$ at the point (0, 1)

b) **[5 marks]** use logarithmic differentiation to find the derivative of the function:
$y=(9x + 7)^3 \sqrt{4x + 5}$

Q–4:

a) **[2×3 marks]** Evaluate the following integrals:
$\int_0^3 x^3 + x - 7\, dx.$

i. $\int_{-1}^{1} \frac{e^x + e^{-x}}{2}\, dx.$

b) **[4marks]** Use integration by substitution to evaluate $\int \frac{x^2}{3-x^3}\, dx$

Q–5:

a) **[5 marks]** Find the area of the region bounded by the graphs of $y=2x^2 - 4x + 6$ *and* $y = -x^2 + 2x + 1$ from x=1 to x=2

b) **[5 marks]** Use integration by parts to evaluate $\int x \cos x\, dx$ dx

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End of questions

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